

6(a)	<i>any two from</i> 1 (MG is an) autoimmune, disease / disorder / condition ; A autoimmunity R autoimmune system 2 (disease in which) failure to distinguish / AW, between self and, non-self ; A between self and, non-self / foreign, antigens 3 (so, at neuromuscular junctions) ref. immune response occurs against / antibody produced against, self- <u>antigens</u> / receptors qualified (see list); A antibodies, bind to / block / attack / AW examples of receptors cell surface muscle cell ACh / neurotransmitter neuromuscular junction postsynaptic 4 <i>idea that</i> defective lymphocytes not destroyed (during development) ;	2
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Question	Answer	Marks
6(b)	<i>max 2 if wrong context</i> <i>e.g. vaccine contains, foreign antigens / antigens of pathogen / pathogen / antibody to bind to lymphocytes</i> <i>any three from</i> 1 vaccine contains antigens ; <i>in context of antigens of defective, T-lymphocytes and B-lymphocytes</i> <i>(accept antibody of defective B-lymphocyte as AW antigen)</i> 2 stimulates an immune response ; 3 detail ; e.g. macrophages and antigen presentation / AW <i>ref. to specific lymphocytes involved</i> clonal selection of lymphocytes recognition by / activation of, lymphocytes clonal expansion of lymphocytes production of plasma cells <i>in context of B-lymphocytes</i> production of memory cells T-helper cells produce cytokine 4 antibody produced against (defective), T-lymphocytes / B-lymphocytes / antibody (that binds ACh receptors) ; <i>in context of primary or secondary response</i> R <i>if context is related to the vaccine containing antibody</i> 5 memory cells, remain / give long-lasting effect / AW ; 6 memory cells allow secondary immune response to (defective) lymphocytes (newly produced) ;	3

Question	Answer	Marks
6(c)	<i>any one from</i> antibiotic resistance, qualified ; A drug for antibiotic e.g. increases risk of antibiotic resistance developing may cause / could lead to problems of, antibiotic resistance an antibiotic resistant bacterial strain may develop R if, immune / immunity, included R antibiotic resistance / mutation, in, TB / disease / people / virus mutation(s) occur in <u>bacteria</u> which, make antibiotics ineffective / prevent antibiotic action ; A mutation occurs which may make bacteria resistant AVP ; e.g. <i>ref. to</i> development of, DR-TB / MDR-TB / XDR-TB <i>(if latent bacteria become active)</i>	1

Question	Answer	Marks
6(d)	<i>max 2 if not in context of the statistic stated (using only Fig. 6.1) mps 2 -8 only</i> <i>points are in context of people who do not know</i> <i>any three from</i> 1 9.2 million do not know that they are, living / infected with, HIV ; ora 27.7 million people know <i>risk to themselves</i> 2 not accessing, ART / treatment ; A only people who know will access ART A if tested, can access ART 3 high(er) viral, loads / levels (compared to many on ART); ora <i>idea of no control over presence of virus in body</i> 4 greater risk of, developing / dying from, <u>HIV/AIDS</u> / HIV-related diseases ; ora A AIDS for HIV / AIDS R HIV <i>context could be, no ART / higher viral loads / no treatment of HIV-related diseases</i> <i>risk to others</i> 5 greater risk of transmitting, virus / HIV ; ora <i>includes placental transmission</i> R HIV/AIDS or AIDS <i>context could be, higher viral load / not practising prevention</i> 6 detail of prevention that may not be practised ; ora e.g. are sharing contaminated needles are donating blood are breastfeeding are not, practising safe sex / using condoms / AW 7 (may develop opportunistic infection so) greater risk of spreading infectious diseases ; <i>consequence for, health care / authorities</i>	3

Question	Answer	Marks
6(d)	8 problems with increase in infection within a population ; ora e.g. increased cost of, care / medication increased pressure on medical staff reduction in size of labour force 9 <i>ref. to</i> testing for HIV ; R testing for HIV/AIDS e.g. indicates more, education / advertising, needed to encourage testing indicates more testing required / focusing on at risk groups 10 AVP ; <i>issues for authorities other than mp 8 and 9</i> e.g. if people do not know status, no contact tracing can occur focus efforts on countries where a high proportion do not know HIV status authorities provide free condoms to try and reduce overall transmission	